

Braid Arrangements for Concurrent Programs

July 28, 2026

Abstract

Precubical sets are a standard tool in the study of concurrent programs. A recent paper by Paliga and Ziemiański showed that the fundamental group of the path space on the terminal precubical set is exactly the Artin braid group. However, they also showed that the induced braid functors for Euclidean sets trivialize. In the world of hyperplane arrangements, this kind of trivialization can be resolved by working in the “complexified” space. We use the same approach to build a new, complexified version of the path space, Ch^* . This agrees with the Salvetti complex of the braid arrangement for the cube, but is defined over any precubical set. By Salvetti’s theorem, then, $\pi_1|\text{Ch}^*(I^n)|$ is the pure braid group. This gives us a new braid invariant for concurrent programs, which has been verified in lean.

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1 Introduction

This work is in 3 parts. First we’ll recap the relevant material in brief. Second, we will perform the construction of the “complexified” path space. Third, we will study some of its properties.

To fix notation, let **Cube** be the box category: its objects are the cubes I^n for $n \geq 0$, and its morphisms are the face embeddings (composites of the coface maps δ_i^ϵ). A *precubical set* is a presheaf on **Cube**; write $\square$ for the category of precubical sets. Concretely, a precubical set K has, for each $n \geq 0$, a set K_n of n -cells, together with face maps

$$d_i^\epsilon : K_n \rightarrow K_{n-1} \quad (1 \leq i \leq n, \epsilon \in \{0, 1\}),$$

where i names a direction and ϵ a side, subject to the precubical identities $d_i^\epsilon d_j^\eta = d_{j-1}^\eta d_i^\epsilon$ for $i < j$. We write I^n both for the standard n -cube as an object of **Cube** and for the presheaf it represents. Also, let $[n]$ be the set $\{0, \dots, n-1\}$.

We usually work with precubical sets carrying designated start and end vertices. A *bipointed* precubical set is a precubical set K with two chosen vertices $\text{init}_K, \text{fin}_K \in K_0$, and morphisms are required to preserve them; write $\square_*$ for the resulting category. Each cube I^n is bipointed by its initial vertex $(0, \dots, 0)$ and final vertex $(1, \dots, 1)$.

Convention 1.1. Named ambient categories are set in bold (**Cat**, **Set**, **Cube**, **Braid**); constructions applied to an object are set upright ($\text{Ch}(K)$, $\text{Run}(K)$, $\text{Sal}(n)$). Functors are written with $\rightarrow$; we reserve $\Rightarrow$ for natural transformations. The standard n -cube is I^n , the symbol $\square$ names the category of precubical sets.

2 Standard build

We recall three constructions on $\square_*$ from Paliga–Ziemiański.

The first is the **wedge** $\vee$. Given $P, Q \in \square_*$, the wedge $P \vee Q$ is the pushout gluing P 's final vertex to Q 's initial vertex:

$$\begin{array}{ccc} I^0 & \xrightarrow{\text{init}_Q} & Q \\ \text{fin}_P \downarrow & \lrcorner & \downarrow \\ P & \longrightarrow & P \vee Q \end{array}$$

The wedge $P \vee Q$ is bipointed by init_P and fin_Q , and $(\square_*, \vee, I^0)$ is monoidal with unit the point I^0 . The **serial wedge** of a list of *positive* naturals is

$$\bigvee[a_1, \dots, a_n] := I^{a_1} \vee \dots \vee I^{a_n},$$

with $\bigvee[] = I^0$.

The second construction is the category of **cube chains**. For $K \in \square_*$, let $\text{Ch}(K)$ be the full subcategory of the slice $\square_*/K$ spanned by the serial wedges: an object is a bipointed map $\bigvee a \rightarrow K$, and a morphism $(\bigvee a \rightarrow K) \rightarrow (\bigvee b \rightarrow K)$ is a bipointed map $\bigvee a \rightarrow \bigvee b$ commuting over K . Such a morphism runs from a finer chain to a coarser one.

The third construction is **refinement**. A *chain* in K is a finite sequence $[c_1, \dots, c_l]$ of cells of K , each of dimension at least 1, such that $\text{fin}(c_i) = \text{init}(c_{i+1})$ for all i , with $\text{init}(c_1) = \text{init}_K$ and $\text{fin}(c_l) = \text{fin}_K$; here $\text{init}(c)$ and $\text{fin}(c)$ denote the images under $c : I^{\dim c} \rightarrow K$ of the initial and final vertices of the cube. A *refinement* replaces a cell c_i by a chain of the cube $I^{\dim c_i}$, post-composed with c_i . Let $\text{Refine}(K)$ be the category of chains, with morphisms generated by refinements and oriented as in $\text{Ch}(K)$ (finer to coarser).

Finally, two properties of K from the Configuration Spaces paper recur. We say K *admits altitude* if there is a function $\text{alt} : K_0 \rightarrow \mathbb{Z}$ with $\text{alt}(\text{fin}(e)) = \text{alt}(\text{init}(e)) + 1$ for every edge $e \in K_1$. We say K is *non-self-linked* (NSL) if every map $I^n \rightarrow K$ is mono.

A few things to note

Lemma 2.1. *The $\vee$ operator is a monoidal product on $\square_*$.*

Lemma 2.2. *$\text{Ch} : \square_* \rightarrow \mathbf{Cat}$ is a lax monoidal functor from $(\square_*, \vee)$ to $(\mathbf{Cat}, \times)$.*

Lemma 2.3. *If K admits altitude and is NSL, then $\text{Ch}(K)$ and $\text{Refine}(K)$ are equivalent, and Ch is strong monoidal there.*

Lemma 2.4. *Serial wedges admit altitudes, and are NSL.*

This is all the foundational material we need.

3 Wedge Machinery

In an effort to keep our proofs better factored, we establish some results here that let us extend constructions on **Cube** to constructions on $\square_*$.

We write E for both extensions below. The variance of P determines which is meant: a covariant P is extended by left Kan extension, a contravariant one by right Kan extension.

Lemma 3.1. *Any functor $P : \mathbf{Cube} \rightarrow \mathbf{Cat}$ extends, by left Kan extension along $\mathbf{Cube} \hookrightarrow \square$, to a functor $E(P) : \square_* \rightarrow \mathbf{Cat}$. If $P(I^0) = \emptyset$, then $E(P)$ is strong monoidal*

$$E(P) : (\square_*, \vee, I^0) \rightarrow (\mathbf{Cat}, \sqcup, \emptyset),$$

and for every serial wedge

$$E(P)(\bigvee a) \simeq \coprod_i P(I^{a_i}).$$

Proof. As a pointwise left Kan extension (a coend over **Cube**), $E(P)$ is cocontinuous, so it sends the wedge pushout $A \leftarrow I^0 \rightarrow B$ to $E(P)(A \vee B) = E(P)(A) \sqcup_{P(I^0)} E(P)(B)$. When $P(I^0) = \emptyset$ this pushout is the coproduct, so the comparison map is an isomorphism; the formula for $\bigvee a$ follows by induction on the number of summands. $\square$

Lemma 3.2. *Any functor $P : \mathbf{Cube}^{\text{op}} \rightarrow \mathbf{Cat}$ extends, by right Kan extension along $\mathbf{Cube} \hookrightarrow \square$, to a $\mathbf{Cat}$ -valued presheaf $E(P) : \square_*^{\text{op}} \rightarrow \mathbf{Cat}$. If $P(I^0)$ is terminal, then $E(P)$ is strong monoidal*

$$E(P) : (\square_*^{\text{op}}, \vee, I^0) \rightarrow (\mathbf{Cat}, \times, \mathbf{1}),$$

and for every serial wedge

$$E(P)(\bigvee a) \simeq \prod_i P(I^{a_i}).$$

Proof. Dually, $E(P)$ is a pointwise right Kan extension, hence continuous, and being contravariant it sends the wedge pushout to the pullback $E(P)(A \vee B) = E(P)(A) \times_{P(I^0)} E(P)(B)$. When $P(I^0)$ is terminal this pullback is the product, so the comparison is an isomorphism; the formula for $\bigvee a$ follows by induction. $\square$

We now instantiate this machinery. The first case is the coordinate functor.

Definition 3.3. Let $\text{Coord} : \mathbf{Cube} \rightarrow \mathbf{Set}$ send the cube I^n to its set of directions $[n] := \{1, \dots, n\}$, and send a face embedding $f : I^n \rightarrow I^m$ to the monotone injection $[n] \hookrightarrow [m]$ that names the n directions f leaves free. Note $\text{Coord}(I^0) = [0] = \emptyset$, so Lemma 3.1 applies to Coord : the extension $E(\text{Coord})$ is strong monoidal, with

$$E(\text{Coord})(\bigvee a) \simeq \prod_i [a_i].$$

For a morphism $f : \bigvee a \rightarrow \bigvee b$ we write $f_* := E(\text{Coord})(f)$ for the induced function on coordinate sets,

$$f_* : \coprod_i [a_i] \longrightarrow \coprod_j [b_j],$$

which records where each coordinate of $\bigvee a$ is sent.

Definition 3.4. Write $\text{blk}(c)$ for the summand index of a coordinate $c \in \coprod_i [a_i]$. By the pushout defining $\bigvee a$, a morphism $f : \bigvee a \rightarrow \bigvee b$ is determined by its restrictions to the summands, and each such restriction is a cell of $\bigvee b$ of positive dimension, hence lies in a single summand: a face embedding

$$f_i : I^{a_i} \longrightarrow I^{b_{\text{blk}(f)(i)}}.$$

This defines $\text{blk}(f)$, and describes f_* on coordinates:

$$f_*(i, d) = (\text{blk}(f)(i), (f_i)_*(d)), \quad d \in [a_i]. \quad (1)$$

In particular $\text{blk} \circ f_* = \text{blk}(f) \circ \text{blk}$.

Lemma 3.5. *For $f : \bigvee a \rightarrow \bigvee b$, the map $\text{blk}(f)$ is monotone and f_* restricts to an order-preserving injection $[a_i] \hookrightarrow [b_{\text{blk}(f)(i)}]$. Consequently $\text{blk} \circ f_*$ is monotone.*

Proof. The restriction of f_* to $[a_i]$ is $(f_i)_*$, which is an order-preserving injection because f_i is a face embedding (Definition 3.3). Monotonicity holds because f preserves the shared vertex fin of summand $i = \text{init}$ of summand $i + 1$. The last claim follows since $\text{blk} \circ f_* = \text{blk}(f) \circ \text{blk}$ and blk is monotone, the order being lexicographic in it. $\square$

Lemma 3.6. *For any morphism $f : \bigvee a \rightarrow I^m$ in $\square$, the function f_* is injective.*

Proof. We recurse on a . If a is empty, then $f_* : [0] \rightarrow [m]$, which is always injective. For the induction case, we have $f : (\bigvee a) \vee I^n \rightarrow I^m$. Now, by the universal property of the pushout, this splits into $h : I^n \rightarrow I^m$ and $g : \bigvee a \rightarrow I^m$. Now, g_* is injective by hypothesis. On the other hand, h is a map of representables, and E is the left Kan extension. So $h_* = \text{Coord}(h)$ is the injective monotone map $[n] \rightarrow [m]$ corresponding to h .

Finally, we show the two images are disjoint. Let $c \in \{0, 1\}^m$ be the image of the glued vertex $\text{fin}(\bigvee a) = \text{init}(I^n)$. Maps into a cube are monotone, raising one coordinate from 0 to 1 per edge. Since h starts at c , so every point in the image of h is reachable from c . So h can only raise coordinates, every direction free in h is 0 at c . On the other hand g ends at c . Every direction g raises has already reached 1 at c . Hence

$$\text{im } h_* \subseteq \{k : c_k = 0\} \quad \text{and} \quad \text{im } g_* \subseteq \{k : c_k = 1\}$$

are disjoint. So $f_* = g_* \sqcup h_*$ is the disjoint union of injective functions with disjoint images. Therefore f_* is injective. $\square$

Lemma 3.7. *For any morphism $f : \bigvee a \rightarrow \bigvee b$ in $\square_*$, the function f_* is bijective.*

Proof. First, let's do injectivity. To begin with, we induct on the length of b , generalizing over a . If b is empty, then $\bigvee b = I^0$, and $\bigvee a$ only has 0-cells. Which means a must also be empty, and $f : I^0 \rightarrow I^0$. And $f_* : [0] \rightarrow [0]$ is always injective.

Now, consider $f : \bigvee a \rightarrow I^m \vee \bigvee b$, our inductive hypothesis ensures for any x , the coordinate map of $\bigvee x \rightarrow \bigvee b$ is injective. By strong monoidality of $\mathbf{Ch}$, f as a morphism of $\mathbf{Ch}(I^m \vee \bigvee b)$ splits into (f_1, f_2) as a morphism in $\mathbf{Ch}(I^m) \times \mathbf{Ch}(\bigvee b)$. Now, by the induction hypothesis, $(f_2)_*$ is injective and $(f_1)_*$ is injective by Lemma 3.6. Then, $f \simeq f_1 \vee f_2$, and $E(\mathbf{Coord})$ is strong monoidal. So

$$f_* = (f_1 \vee f_2)_* = (f_1)_* \sqcup (f_2)_*$$

which is the disjoint union of functions with disjoint codomains. Therefore f_* is injective.

For surjectivity, we take a counting argument. Since f is bipointed it preserves altitude, and the altitude of the final vertex of $\bigvee a$ is $\sum_i a_i$. So $\sum_i a_i = \sum_j b_j$, and f_* is a cardinality-preserving injection, hence bijective. $\square$

4 The Complexification Construction

This observation comes from a simple presentation morse flow-like information on the cube. Associate each coordiante of the n -cube with a real nubmer. So coordinates of $\mathbb{R}^n$ behave like a kind of execution information: when did each event occur. Then we break $\mathbb{R}^n$ into regions corresponding to wedge maps: sets of equal coordinates give coarser wedge maps. This turns out to correspond to the braid arrangement of plane.

We define the braid arrangement and its related constructions following Paris. But we specialize it to braid arrangement for convenience here. The braid arrangement A_{n-1} is the family of hyperplanes

$$H_{ij} = \{x \in \mathbb{R}^n : x_i = x_j\}, \quad 1 \leq i < j \leq n,$$

one for each pair of coordinates. Orient H_{ij} by the sign of $x_i - x_j$. Each point $x \in \mathbb{R}^n$ then assigns to every hyperplane a sign in $\{-, 0, +\}$, and the sign vectors so obtained are the **covectors**, or faces, of the arrangement:

$$\text{Face}(n) := \{F : \{(i, j) : 1 \leq i < j \leq n\} \rightarrow \{-, 0, +\} \mid F \text{ arises from some } x \in \mathbb{R}^n\}.$$

A covector records the weak order of the coordinates of x , so it is the same data as an *ordered set partition* of $\{1, \dots, n\}$: the blocks are the coordinates of equal value, listed in increasing order of that value. We order covectors by

$$F \leq G \quad :\Longleftrightarrow \quad \forall h, F(h) = 0 \text{ or } F(h) = G(h),$$

so that G resolves some of the ties left open by F . The maximal covectors — those never taking the value 0 — are the **topes**. A tope has all coordinates distinct, hence is a total order on $\{1, \dots, n\}$; there are exactly $n!$ of them, one per permutation.

Sadly, its order complex is contractible. Salvetti's construction reruns the story over $\mathbb{C}$ and finds much richer homotopy. Define the **Salvetti poset**

$$\text{Sal}(n) := \{(F, C) \mid F \in \text{Face}(n), C \text{ a tope}, F \leq C\},$$

whose order uses the composition of covectors,

$$(F \circ G)(h) := \begin{cases} F(h) & \text{if } F(h) \neq 0, \\ G(h) & \text{if } F(h) = 0, \end{cases}$$

namely

$$(F, C) \preceq (F', C') \quad :\Longleftrightarrow \quad F \leq F' \quad \text{and} \quad C' = F' \circ C.$$

By Salvetti's theorem $|\text{Sal}(n)|$ is homotopy equivalent to the complement of the complexified braid arrangement in $\mathbb{C}^n$ — the ordered configuration space of n points in $\mathbb{C}$. So $\pi_1|\text{Sal}(n)|$ is the pure braid group P_n . Intuitively, each face is annotated with a compatible tope, a local orientation; crossing a face can flip the orientation, and these flips are the braid letters. The ‘Configuration Spaces’ construction retains the same orientation data, breaking every tie at a face in all possible ways.

This has an alternative packaging as a Grothendieck construction, which is what we will imitate. Let

$$\text{Tope} : \text{Face}(n) \rightarrow \mathbf{Set}, \quad \text{Tope}(F) := \{C \text{ a tope} : F \leq C\},$$

be the covariant functor sending $F \leq F'$ to the map $C \mapsto F' \circ C$ (composing on the left by F' carries a tope above F to one above F'). Then

$$\text{Sal}(n) \cong \int \text{Tope},$$

the Grothendieck construction: its objects are the pairs (F, C) with $C \in \text{Tope}(F)$, and its morphisms are exactly the relations $\preceq$ above.

To rebuild this generically on the precubical side, let's start with the corresponding definition of tope.

Definition 4.1. A serial wedge is **one dimensional** if all its entries are 1.

Definition 4.2. Let $\text{Run}(K)$ be the full subcategory of $\text{Ch}(K)$ whose objects have one dimensional serial wedges as domain.

Lemma 4.3. *Being one dimensional is preserved under $\vee$, so Run is a monoidal functor. In particular $\mathbf{List} \mathbb{N}^+ \rightarrow \text{Run}(\vee -)$ is a lax monoidal functor.*

Now, the morphisms we want to build are a bit unusual, because they are built from degeneracies, which are not morphisms of $\square$. That is, we want to build a contravariant map from $\text{Ch}(K)$ to runs, mirroring the order from Salvetti:

$$(\bigvee a \rightarrow \bigvee b) \longmapsto (\text{Run}(\bigvee b) \rightarrow \text{Run}(\bigvee a)).$$

To achieve this we use the cell-list presentation of a run. Let $\text{Edge}(K)$ be the set of chains in $\text{Refine}(K)$ all of whose cells are one dimensional. Serial wedges are NSL and admit altitude, so $\text{Ch}(K) \simeq \text{Refine}(K)$ there, and restricting that equivalence to the one dimensional part gives a bijection

$$\text{Run}(K) \cong \text{Edge}(K)$$

for every serial wedge K : a one dimensional wedge over K is the same data as its list of edges.

Now, given a face embedding $f : I^n \rightarrow I^m$, it defines a unique projection of cells of I^m to cells of I^n , non-increasing on dimension: it forgets the directions of I^m that f does not leave free. Call it $\hat{f}$. So we define

$$\rho(f) : \text{Edge}(I^m) \rightarrow \text{Edge}(I^n), \quad [c_1, \dots, c_k] \mapsto \text{norm}[\hat{f}(c_1), \dots, \hat{f}(c_k)],$$

where norm removes all the 0-cells, and transport it along $\text{Run} \cong \text{Edge}$ to get

$$\rho(f) : \text{Run}(I^m) \rightarrow \text{Run}(I^n).$$

Concretely, a run of I^m is a bijection $r_* : [m] \rightarrow [m]$ (Lemma 3.7), and $\rho(f)$ restricts it along f_* : the run $\rho(f)(r)$ is the unique one satisfying

$$f_* \circ \rho(f)(r)_* = r_* \circ \iota \quad \text{for some strictly increasing } \iota : [n] \rightarrow [m], \quad (2)$$

that is, it visits the directions left free by f in the order r meets them. Note that the result is indeed a one dimensional chain in I^n because

1. $\hat{f}$ never increases dimension, so all dimensions are less than or equal to 1;
2. normalization removes all the 0-cells, so every cell is one dimensional;
3. every projection respects initial/final points.

Lemma 4.4. $\rho : \mathbf{Cube}^{\text{op}} \rightarrow \mathbf{Set}$ is a functor, and $\rho(I^0)$ is a singleton.

Proof. The proof of functoriality mostly just exploits the fact that cubes are NSL and admit altitudes, so that $\text{Run}(I^n) \cong \text{Edge}(I^n)$ as sets. First, ρ preserves identities, as the projection for the identity embedding is also the identity. Second, if $f : I^n \rightarrow I^m$ and $g : I^m \rightarrow I^l$ are face embeddings, then their corresponding projections respect composition.

For the singleton, observe that $\rho(I^0) = \text{Run}(I^0)$, but the only one dimensional serial wedge which maps into I^0 is the empty one. So indeed $\rho(I^0)$ is a singleton. $\square$

That means two things. Firstly, it means ρ is itself a precubical set. Second, it means our lifting machinery sends ρ to $E(\rho) : \square_*^{\text{op}} \rightarrow \mathbf{Cat}$. Lastly, let $\square^\vee$ be the full subcategory of serial wedges in $\square_*$. There is also a forgetful functor $W(K) : \text{Ch}(K) \rightarrow \square^\vee$ sending $(\bigvee a \rightarrow K) \mapsto \bigvee a$. Then we can define our key functor

$$\text{Lines}(K) := E(\rho) \circ W(K)^{\text{op}}$$

which is a presheaf on $\text{Ch}(K)$. Then we define

$$\text{Ch}^*(K) := \int \text{Lines}(K)$$

the category of elements of the presheaf $\text{Lines}(K)$. Defining Ch^* abstractly has some conveniences later. Let's unpack the definition of Ch^* to have a more concrete presentation of it.

Objects of $\text{Ch}^*(K)$ are pairs

$$(\bigvee a \rightarrow K, x \in \text{Run}(\bigvee a)), \text{ i.e. } \bigvee[1, \dots, 1] \rightarrow \bigvee a \rightarrow K.$$

The intuition here matches the Salvetti story precisely: we have a wedge map into K , coupled with orientation info about it. Morphisms are a little uglier. A morphism

$$(\bigvee a \rightarrow K, x \in \text{Run}(\bigvee a)) \longrightarrow (\bigvee b \rightarrow K, y \in \text{Run}(\bigvee b))$$

unpacks to

1. A morphism $f : (\bigvee a \rightarrow K) \rightarrow (\bigvee b \rightarrow K)$, i.e. a bipointed map $f' : \bigvee a \rightarrow \bigvee b$ commuting over K .
2. The condition $x = \text{Lines}(f)(y)$: we take the run y , project it along f' , and require the result to be x . Since the fibers of $E(\rho)$ are discrete, this is a condition, not extra data — x is determined by f' and y .

The diagram below records the story. The solid arrows are morphisms of $\text{Ch}^*(K)$; the dashed $\sigma(f)$ (a permutation of positions) and $E(\rho)(f)$ (the run projection) are plain functions, not morphisms in $\square_*$, and they point oppositely — $E(\rho)(f)$ carries y back to x , and $\sigma(f)$ is the induced reindexing.

$$\begin{array}{ccccc}
 \vee[1, \dots, 1] & \xrightarrow{x} & \vee a & & \\
 \downarrow \sigma(f) & & \downarrow f & \searrow & \\
 \vee[1, \dots, 1] & \xrightarrow{y} & \vee b & \xrightarrow{\quad} & K
 \end{array}
 \quad
 \begin{array}{c}
 \text{---} E(\rho)(f) \text{---} \\
 \text{---} \text{---} \text{---}
 \end{array}$$

Let's look more closely at the construction of σ . Note that a run $r \in \text{Run}(\vee p)$ is itself a bipointed map $\vee[1, \dots, 1] \rightarrow \vee p$, so Definition 3.3 already applies to it, and by Lemma 3.7 its coordinate map

$$r_* : [n] \xrightarrow{\sim} \coprod_i [p_i]$$

is exactly a linear ordering of the coordinates of $\vee p$. Conjugating the coordinate bijection $f'_* : \coprod_i [a_i] \rightarrow \coprod_j [b_j]$ by the two run orderings gives a permutation of $[n]$,

$$\sigma(f) := y_*^{-1} \circ f'_* \circ x_* \in \Sigma_n,$$

equivalently $y_* \circ \sigma(f) = f'_* \circ x_*$. This permutation $\sigma(f)$ is what will power our braid functor.

Lemma 4.5. $\sigma : \text{Ch}^*(K) \rightarrow \bigsqcup_n \Sigma_n$ is a functor.

Proof. Along a composite the middle ordering y_* cancels,

$$\sigma(g \circ f) = z_*^{-1} \circ g'_* \circ f'_* \circ x_* = (z_*^{-1} \circ g'_* \circ y_*)(y_*^{-1} \circ f'_* \circ x_*) = \sigma(g) \sigma(f),$$

and

$$\sigma(\text{id}) = x_*^{-1} \circ (\text{id})_* \circ x_* = \text{id},$$

□

Lemma 4.6. Let $f : (\vee a, x) \rightarrow (\vee b, y)$ in $\text{Ch}^*(K)$ and suppose $\text{blk}(x_*(k)) = \text{blk}(x_*(l))$. Then $k < l$ if and only if $\sigma(f)(k) < \sigma(f)(l)$.

Proof. By Lemma 3.2, $\text{Run}(\vee a) \cong \prod_i \text{Run}(I^{a_i})$. So $x = x_1 \vee \dots \vee x_n$ and $(x_i)_* : [a_i] \rightarrow [a_i]$. Since $\text{blk} \circ x_*$ is monotone (Lemma 3.5), and both blk and $(\cdot)_*$ are monoidal, we can define $s_i^a := \sum_{i' < i} a_{i'}$ and we the decomposition

$$x_*(s_i^a + t) = (i, (x_i)_*(t)), \quad t \in [a_i]. \quad (3)$$

Put $i := \text{blk}(x_*(k)) = \text{blk}(x_*(l))$ and $j := \text{blk}(f'_*)(i)$, and write $k = s_i^a + t$ and $l = s_i^a + t'$. Componentwise the morphism condition $x = E(\rho)(f')(y)$ reads $x_i = \rho(f_i)(y_j)$, so (2) gives $(f_i)_* \circ (x_i)_* = (y_j)_* \circ \iota$ with ι strictly increasing. Hence

$$y_*(\sigma(f)(k)) = f'_*(x_*(s_i^a + t)) = f'_*(i, (x_i)_*(t)) = (j, (f_i)_*((x_i)_*(t))) = (j, (y_j)_*(\iota(t))) = y_*(s_j^b + \iota(t)),$$

As y_* is injective, $\sigma(f)(k) = s_j^b + \iota(t)$ and $\sigma(f)(l) = s_j^b + \iota(t')$; since ι is strictly increasing, $\sigma(f)(k) < \sigma(f)(l)$ iff $t < t'$ iff $k < l$. □

5 The Braiding

You'll notice that the K side is not involved in the computation of the permutation, or the proofs of its functoriality.

This comes from a rather nice fact about Grothendieck constructions:

$$\begin{array}{ccc}
 \mathrm{Ch}^*(K) = \int (E(\rho) \circ W(K)^{\mathrm{op}}) & \xrightarrow{Q} & \int E(\rho) = \square^\vee / \rho \\
 \pi \downarrow & \lrcorner & \downarrow \pi \\
 \mathrm{Ch}(K) & \xrightarrow{W(K)} & \square^\vee
 \end{array}$$

The map Q is just a projection to the first arrow of $\vee[1, \dots, 1] \rightarrow \vee a \rightarrow K$. And ρ is the classifier for runs. Since E is essentially just the Yoneda embedding, the right corner turns into a slice category of wedges over ρ . Secondly, K does not appear on the right hand side.

Lemma 5.1. *The functor Q is always faithful.*

Proof. Let $\vee[1, \dots, 1] \rightarrow \vee a \rightarrow K$ and $\vee[1, \dots, 1] \rightarrow \vee b \rightarrow K$ be two objects of $\mathrm{Ch}^*(K)$. Two morphisms f, g between these objects are equal exactly when they have the same underlying map $\vee a \rightarrow \vee b$, which happens if and only if $Q(f) = Q(g)$. So Q is indeed faithful. $\square$

But, we can do a nicer trick

Lemma 5.2. *Let Z be the terminal precubical set. Then $\mathrm{Ch}^*(Z)$ is isomorphic to $\square^\vee / \rho$.*

Proof. We know Q is faithful already. To show Q is full, again we take two objects $\vee[1, \dots, 1] \rightarrow \vee a \rightarrow Z$ and $\vee[1, \dots, 1] \rightarrow \vee b \rightarrow Z$. Let f be a morphism in $\square^\vee / \rho$. That is, $f : \vee a \rightarrow \vee b$ forming a commuting triangle:

$$\begin{array}{ccc}
 \vee a & \xrightarrow{\quad} & \rho \\
 f \downarrow & \nearrow & \\
 \vee b & &
 \end{array}$$

But, since Z is the terminal precubical set, every map commutes in the triangle

$$\begin{array}{ccc}
 \vee a & \xrightarrow{\quad} & Z \\
 f \downarrow & \nearrow & \\
 \vee b & &
 \end{array}$$

So Q is full.

Surjectivity on objects is equally straightforward. Every wedge $\vee a \rightarrow \rho$ is equivalent to a $\vee[1, \dots, 1] \rightarrow \vee a$. And every $\vee a$ has a unique map $\vee a \rightarrow Z$. So we have a corresponding object of $\mathrm{Ch}^*(Z)$. $\square$

But we need a device for building braid words from permutations, which brings us to the Garside germ presentation of braids. For $\sigma \in \Sigma_n$ write

$$\mathrm{Inv}(\sigma) := \{ (i, j) \in [n] \times [n] \mid i < j \text{ and } \sigma(j) < \sigma(i) \}$$

for its set of *inversions*, and $\ell(\sigma) := |\text{Inv}(\sigma)|$ for its *length*. The germ presentation takes one generator $[\sigma]$ for each $\sigma \in \Sigma_n$, subject to

$$[\sigma][\sigma'] = [\sigma\sigma'] \quad \text{whenever} \quad \ell(\sigma\sigma') = \ell(\sigma) + \ell(\sigma').$$

The key fact is a “no double crossings” feature of Ch^* . That is, once you refine a coordinate, you can’t refine it again. Let’s first state this in pure combinatorics.

Lemma 5.3. *Let f and g be permutations of $[n]$, and suppose that $\text{Inv}(f) \subseteq \text{Inv}(g \circ f)$. Then $\ell(g \circ f) = \ell(g) + \ell(f)$.*

Proof. Every element of $\text{Inv}(g \circ f)$ either lies in $\text{Inv}(f)$ or not, so

$$\text{Inv}(g \circ f) = (\text{Inv}(g \circ f) \cap \text{Inv}(f)) \sqcup (\text{Inv}(g \circ f) \setminus \text{Inv}(f)).$$

By hypothesis $\text{Inv}(f) \subseteq \text{Inv}(g \circ f)$, so the first set equals $\text{Inv}(f)$ and contributes $\ell(f)$.

For the second, a pair $(k, l) \notin \text{Inv}(f)$ has $f(k) < f(l)$, and then $(k, l) \in \text{Inv}(g \circ f)$ iff $g(f(k)) > g(f(l))$, i.e. iff $(f(k), f(l)) \in \text{Inv}(g)$. Since f is a bijection, $(k, l) \mapsto (f(k), f(l))$ restricts to a bijection

$$\text{Inv}(g \circ f) \setminus \text{Inv}(f) \xrightarrow{\sim} \{(p, q) \in \text{Inv}(g) : f^{-1}(p) < f^{-1}(q)\}.$$

This target is all of $\text{Inv}(g)$: otherwise take $(p, q) \in \text{Inv}(g)$ and set $k = f^{-1}(q) < l = f^{-1}(p)$; then $f(k) = q > p = f(l)$ gives $(k, l) \in \text{Inv}(f)$, while $g(f(k)) = g(q) < g(p) = g(f(l))$ gives $(k, l) \notin \text{Inv}(g \circ f)$, contradicting $\text{Inv}(f) \subseteq \text{Inv}(g \circ f)$. Hence the second set contributes $\ell(g)$, and $\ell(g \circ f) = \ell(f) + \ell(g)$. $\square$

Lemma 5.4. *Let $g : \bigvee a \rightarrow \bigvee b$, $f : \bigvee b \rightarrow \bigvee c$, and $S := \{(k, l) \mid \text{blk}(g_*(k)) = \text{blk}(g_*(l))\}$. Then $\text{Inv}(g_*) = \text{Inv}(f_* \circ g_*) \cap S$; in particular $\text{Inv}(g_*) \subseteq \text{Inv}(f_* \circ g_*)$.*

Proof. If $k < l$ then $\text{blk}(g_*(k)) \leq \text{blk}(g_*(l))$ by Lemma 3.5, and if that is strict then $g_*(k) < g_*(l)$; so $\text{Inv}(g_*) \subseteq S$. On S the two coordinates share a summand of $\bigvee b$, where f_* is an order-preserving injection, so $(k, l) \in \text{Inv}(g_*)$ iff $(k, l) \in \text{Inv}(f_* \circ g_*)$. Hence $\text{Inv}(f_* \circ g_*) \cap S = \text{Inv}(g_*) \cap S = \text{Inv}(g_*)$. $\square$

Lemma 5.5. *The map $f \mapsto [\sigma(f)]$ yields a functor $\text{Conc} : \text{Ch}^*(K) \rightarrow \mathbf{Braid}$.*

Proof. An object goes to the braid object on its number of coordinates, and $\sigma(\text{id}) = \text{id}$. For composition we need $[\sigma(f)][\sigma(g)] = [\sigma(f) \circ \sigma(g)]$, which by the germ presentation holds once $\ell(\sigma(f) \circ \sigma(g)) = \ell(\sigma(f)) + \ell(\sigma(g))$, and by Lemma 5.3 it is enough to prove $\text{Inv}(\sigma(g)) \subseteq \text{Inv}(\sigma(f) \circ \sigma(g))$ for composable $g : (\bigvee c, r) \rightarrow (\bigvee b, q)$ and $f : (\bigvee b, q) \rightarrow (\bigvee a, p)$.

Let $(k, l) \in \text{Inv}(\sigma(g))$ and put $u := \sigma(g)(k)$, $v := \sigma(g)(l)$, so $k < l$ and $v < u$. Since $q_* \circ \sigma(g) = g'_* \circ r_*$, we have $q_*(u) = g'_*(r_*(k))$ and $q_*(v) = g'_*(r_*(l))$, so

$$\text{blk}(q_*(u)) = \text{blk}(g'_*)(\text{blk}(r_*(k))) \leq \text{blk}(g'_*)(\text{blk}(r_*(l))) = \text{blk}(q_*(v)),$$

using $k < l$ and Lemma 3.5 twice. On the other hand $v < u$ gives $\text{blk}(q_*(v)) \leq \text{blk}(q_*(u))$. So $\text{blk}(q_*(u)) = \text{blk}(q_*(v))$: the two positions lie in a common summand of $\bigvee b$. Now Lemma 4.6 applied to f turns $v < u$ into $\sigma(f)(v) < \sigma(f)(u)$, which is exactly $(k, l) \in \text{Inv}(\sigma(f) \circ \sigma(g))$. $\square$

This looks a bit silly, because all we've done is proven our braid words have no cancellation. But that only means we found a nice basis. And in fact, it means we're working in the positive braid monoid. We recover the full braiding structure by lifting to the free groupoid, which is now free.

Lemma 5.6. *For any precubical set K , if there is an $f : I^2 \rightarrow K$, then $\text{Conc}(K)$ is nontrivial.*

Proof. It suffices to find a morphism in $\text{Ch}^*(K)$ whose permutation is not the identity. Consider the map $p : \mathbb{V}[1, 1] \rightarrow \mathbb{V}[2]$ going along the bottom edge, then the right edge, and the map $q : \mathbb{V}[1, 1] \rightarrow \mathbb{V}[2]$ going along the top edge. We have

$$\begin{array}{ccccc}
 \mathbb{V}[1, 1] & \xrightarrow{\text{id}} & \mathbb{V}[1, 1] & & \\
 \sigma(p) \downarrow \text{dashed} & & \downarrow p & \searrow & \\
 \mathbb{V}[1, 1] & \xrightarrow{q} & \mathbb{V}[2] & \xrightarrow{f} & K
 \end{array}$$

where $\sigma(p) = \{1 \mapsto 2, 2 \mapsto 1\}$ has an inversion, so $[\sigma(p)]$ is not the identity braid. □

Lemma 5.7. *The category $\text{Conc}(I^n)$ is isomorphic to the pure braids.*

6 Additional Facts

A few facts that have somewhat nasty proofs, but are mechanical

1. The analogy holds for the cube: $\text{Sal}(n) \simeq \text{Ch}^*(I^n)$. The proof starts by proving $\text{Face}(n) \simeq \text{Ch}(I^n)$ via ordered set partitions. Then the observation that runs and topes agree, and one can brute force it from there. This is done in the lean repo.
2. In fact, not only does that mean $\pi_1|\text{Ch}^*(I^n)|$ is the pure braid group, it comes from the lifting of Conc to the free groupoid on $\text{Ch}^*(I^n)$.
3. The discrete opfibration $\text{Ch}^*(K) \rightarrow \text{Ch}(K)$ does not have a canonical way to pick base points in general.